

The diagram illustrates the sulfur cycle and the role of DsrMKJOP. Key components include:

- Sulfur Species:** HS^- , $\text{S}(0)$, SO_3^{2-} , $\text{S}_2\text{O}_3^{2-}$, $\text{S}_4\text{O}_6^{2-}$, SO_3^{2-} , SO_4^{2-} , and **APS**.
- Enzymes and Pathways:**
 - PhsABC:** Catalyzes the conversion of $\text{S}_2\text{O}_3^{2-}$ to HS^- and SO_3^{2-} .
 - SoxYZXAB:** Catalyzes the conversion of $\text{S}_2\text{O}_3^{2-}$ to SO_4^{2-} .
 - SoxYZXABCD:** Catalyzes the conversion of $\text{S}_4\text{O}_6^{2-}$ to SO_4^{2-} .
 - TetH:** Catalyzes the conversion of $\text{S}_4\text{O}_6^{2-}$ to SO_3^{2-} .
 - SoxB:** Catalyzes the conversion of $\text{S}_4\text{O}_6^{2-}$ to SO_3^{2-} .
 - TsdA/DoxDA:** Catalyzes the conversion of $\text{S}_2\text{O}_3^{2-}$ to $\text{S}_4\text{O}_6^{2-}$.
 - Ttr/Otr:** Catalyzes the conversion of $\text{S}_4\text{O}_6^{2-}$ to $\text{S}_2\text{O}_3^{2-}$.
 - FccAB/Sqr:** Catalyzes the conversion of HS^- to $\text{S}(0)$.
 - SOR:** Catalyzes the conversion of $\text{S}(0)$ to SO_3^{2-} .
 - sHdrAB1B2B3C1C2, DsrABCEFH, Sdo:** Catalyze the conversion of SO_3^{2-} to $\text{S}(0)$.
 - SoeABC, SorAB, SorT:** Catalyze the conversion of SO_3^{2-} to SO_4^{2-} .
 - Sat:** Catalyzes the conversion of SO_4^{2-} to **APS**.
 - AprAB:** Catalyzes the conversion of **APS** to SO_3^{2-} .
- DsrMKJOP:** A central enzyme complex involved in the conversion of HS^- to $\text{S}(0)$ and SO_3^{2-} to $\text{S}(0)$, as indicated by the dotted arrows and the label e^- .

[illegible]

Dissimilatory sulfide oxidation pathway

The diagram illustrates the dissimilatory sulfide oxidation pathway across three compartments: Periplasm, Membrane, and Cytoplasm.

- Periplasm:**
 - FccAB** (orange box) is involved in a cycle with HS^- and HS_n^- .
 - Sqr** (green box) is located in the Membrane, converting $\text{MQ} \rightarrow \text{MQH}_2$.
- Membrane:**
 - DsrC** (blue circle) is located in the Membrane, converting $\text{MQ} \rightarrow \text{MQH}_2$.
 - DsrC** (blue circle) is also involved in a cycle with S^0 and $\text{S}^{(0)}$.
- Cytoplasm:**
 - DsrEFH** (orange box) is involved in a cycle with HS^- and HS-SH .
 - DsrBA** (yellow box) and **DsrL** (yellow box) are involved in the conversion of SO_3^{2-} to APS .
 - AprBA** (yellow box) is located in the Membrane, converting APS to SO_4^{2-} .
 - Sat** (yellow box) is located in the Cytoplasm, converting SO_4^{2-} to SO_4^{2-} .

The pathway is labeled **Dissimilatory sulfide oxidation pathway**.